

FINANCIAL PERFORMANCE ANALYSIS

Retail Sales Data Analysis & Business Insights

A Personal Data Analytics Project

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Tools & Technologies

Microsoft Excel • SQL (PostgreSQL) • Power BI

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1. PROJECT OVERVIEW

Financial Performance Analysis is a data analytics project developed using **Microsoft Excel, PostgreSQL/SQL, and Microsoft Power BI** to analyze a Superstore sample transaction dataset.

The project focuses on understanding business performance through **sales, profit, quantity, orders, product, category, regional, segment, and monthly sales analysis**.

The data was first prepared and analyzed using Excel, followed by SQL-based business analysis. The results were then transformed into an interactive Power BI dashboard containing key performance indicators, analytical charts, and filters.

The project demonstrates an end-to-end data analytics workflow, from **data preparation and querying to visualization and business insight generation**.

2. PROJECT OBJECTIVE

The main objectives of this project are:

- To clean and prepare transaction data for analysis.
- To perform initial data analysis using Microsoft Excel.
- To use SQL queries for extracting meaningful business insights.
- To calculate and analyze important performance metrics such as sales, profit, quantity, and orders.
- To identify high-performing products and categories.
- To analyze sales performance across different regions and customer segments.
- To examine monthly sales trends.
- To develop an interactive Power BI dashboard for business reporting.
- To present analytical findings in a clear and professional format.

3. DATASET DESCRIPTION

- **Dataset:** Superstore Sample Dataset
- The dataset contains **transaction-level retail data** covering information related to orders, customers, products, sales, profit, quantity, categories, regions, and dates.
- The dataset was used as a **sample dataset for analytical and portfolio purposes** and does not represent live company transactions.
- Major fields used in the analysis

Field	Description
Order Date	Date on which the order was placed
Customer Name	Customer associated with the order
Region	Geographical region
State	Customer/order state
Category	Product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Sales/revenue generated
Profit	Profit generated
Quantity	Number of units sold
Segment	Customer segment

4. TOOLS & TECHNOLOGIES

Microsoft Excel

Used for:

- Data cleaning
- Data preparation
- Pivot table analysis
- Initial KPI calculations
- Preliminary exploration

PostgreSQL / SQL

Used for:

- Data querying
- Filtering and sorting
- Aggregation
- Grouping

- Business analysis
- Extracting meaningful insights

Microsoft Power BI

Used for:

- Data visualization
- KPI cards
- Interactive charts
- Slicers and filters
- Dashboard development
- Business reporting

DAX

Used in Power BI for creating measures such as:

Total Sales

Total Sales = SUM(Orders[Sales])

5. PROJECT WORKFLOW

The project followed an end-to-end analytics workflow:

Superstore Sample Dataset

↓

Data Cleaning & Preparation — Excel

↓

Business Analysis — SQL

↓

Data Visualization & Dashboard — Power BI

↓

Insights & Recommendations

6. DATA CLEANING AND EXCEL ANALYSIS

Microsoft Excel was used as the first stage of the project to prepare the Superstore sample dataset and perform preliminary analysis. This stage helped ensure that the data was suitable for further analysis using SQL and Power BI.

6.1 Data Preparation

The dataset was initially reviewed to understand its structure, columns, records, and data types. The data was organized into a structured format so that it could be used consistently across Excel, SQL, and Power BI.

The main preparation activities included:

- Reviewing the available columns and records.
- Checking the structure of the dataset.
- Converting the **Order Date** column into the appropriate date format.
- Checking numerical fields such as Sales, Profit, and Quantity.
- Identifying and reviewing potential data inconsistencies.
- Preparing the cleaned dataset for SQL analysis.

6.2 Pivot Table Analysis

Pivot Tables were created in Excel to perform an initial analysis of the dataset.

The analysis included:

- Sales by region.
- Sales by category.
- Sales and profit analysis.
- Product-level analysis.
- Customer and regional performance.
- Monthly and yearly sales analysis.

Pivot Tables helped summarize a large number of transaction records and provided an initial understanding of the business performance before moving to SQL and Power BI.

	A	B	C	D
1				
2				
3	Row Labels	Sum of Sales	Sum of Profit	
4	Furniture	7,54,748	19,730	
5	Office Supplies	7,31,893	1,26,023	
6	Technology	8,39,893	1,46,543	
7	Grand Total	23,26,534	2,92,297	
8				
9				

Excel Pivot Table – Sales and Profit by Category

6.3 KPI Calculation

Key performance indicators were calculated to understand the overall performance of the dataset.

The major KPIs considered were:

- **Total Sales**

- **Total Profit**
- **Total Quantity**
- **Total Orders**
- **Average Sales per Order**

These KPIs were later used as the foundation for the Power BI dashboard.

6.4 Initial Findings from Excel

The Excel analysis provided an initial understanding of sales and profitability patterns across different business dimensions.

The analysis showed that performance varied across:

- Regions
- States
- Product categories
- Products
- Customer segments
- Time periods

The Excel stage therefore served as the foundation for the more detailed SQL analysis and interactive Power BI reporting.

7. SQL DATA ANALYSIS

After preparing the dataset in Excel, the cleaned data was exported as a CSV file and imported into PostgreSQL for SQL-based analysis.

A database named **superstore_db** was created, and an **orders** table was used to store the cleaned transaction data.

7.1 SQL Operations Performed

The following SQL concepts were used during the analysis:

1. SELECT
2. WHERE
3. ORDER BY
4. GROUP BY
5. HAVING
6. Aggregate functions such as SUM(), COUNT(), AVG(), and MAX()
7. JOINS
8. CASE statements
10. Subqueries

9. CTEs (Common Table Expressions)

10. Window functions

11. These operations were used to filter, summarize, compare, and analyze the transaction dataset and extract meaningful business insights.

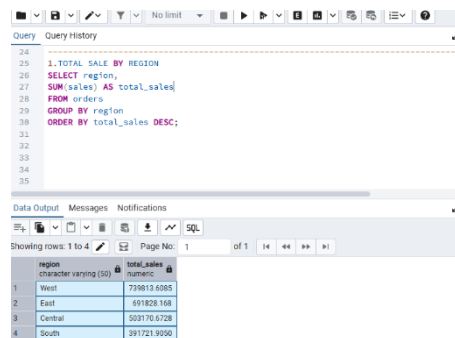
SQL code

```
SELECT Region, SUM(Sales) AS Total_Sales
```

```
FROM Orders
```

```
GROUP BY Region
```

```
ORDER BY Total_Sales DESC;
```



The screenshot shows a SQL query editor with a query history pane on the left. The query is as follows:

```
1. TOTAL SALE BY REGION
SELECT region,
SUM(sales) AS total_sales
FROM orders
GROUP BY region
ORDER BY total_sales DESC;
```

The output is displayed in a table with the following data:

region	total_sales
West	729813.6965
East	591828.168
Central	503170.6728
South	391721.9950

SQL Query and Regional Sales Analysis Output

7.2 Business Analysis Using SQL

SQL queries were used to analyze:

- Total sales and profit.
- Sales by region.
- Sales by state.
- Sales by category.
- Product performance.
- Customer performance.
- Quantity sold.
- Average sales.
- Highest and lowest performance values.

GROUP BY was particularly useful for comparing performance across categories, regions, states, and other business dimensions, while HAVING was used to filter aggregated results.

7.3 SQL Analysis Outcome

The SQL analysis provided structured business insights from the transaction-level data and helped identify patterns that could later be represented visually in Power BI.

SQL also demonstrated the ability to work with a relational database and perform analytical queries rather than relying only on spreadsheet-based analysis.

8. POWER BI DASHBOARD DEVELOPMENT

Microsoft Power BI was used to transform the analyzed Superstore data into an interactive and visually engaging business dashboard. The dashboard was designed to provide a consolidated view of sales and profitability performance and allow users to explore the data using interactive filters.

8.1 Data Connection and Preparation

The cleaned dataset was imported into Power BI and reviewed to ensure that the fields were correctly recognized and suitable for visualization.

The **Order Date** field was configured as a date field so that sales trends could be analyzed over time.

DAX measures were created to calculate important performance indicators.

Total Sales

Total Sales = SUM(Orders[Sales])

Similar measures were used to support the dashboard's key performance indicators.

8.2 Key Performance Indicators

Four major KPI cards were created at the top of the dashboard to provide an immediate overview of business performance:

- **Total Sales** – Represents the overall sales generated.
- **Total Profit** – Represents the total profit generated.
- **Total Quantity** – Represents the total number of units sold.
- **Total Orders** – Represents the number of orders in the dataset.

These KPIs allow users to understand the overall performance before examining individual charts.

8.3 Sales and Profit by Category

A category-level visual was created to compare **Sales and Profit across product categories**.

This visualization helps identify:

- Categories generating higher sales.
- Categories generating higher profit.
- Differences between sales performance and profitability.

This comparison provides a useful business perspective because a category with high sales does not necessarily have the highest profitability.

8.4 Top 10 Products by Sales

A **Top 10 Products by Sales** visualization was created to identify the products contributing the highest sales.

This analysis helps users understand:

- Which products are major contributors to revenue.
- Which products have stronger sales performance.
- Where product-level performance is concentrated.

The visual was limited to the top 10 products to keep the dashboard focused and easy to interpret.

8.5 Monthly Sales Trend

A monthly sales trend visualization was created to analyze changes in sales over time.

The Order Date hierarchy was used to organize the data at the **month level**.

The visualization helps identify:

- Monthly sales patterns.
- Periods of relatively higher or lower sales.
- Changes in sales performance over the selected year.

The dashboard also includes a **Year slicer**, allowing users to select a year and observe the corresponding monthly sales trend.

8.6 Regional Sales Analysis

A regional analysis visual was included to compare sales performance across different geographical regions.

The analysis allows users to understand how sales are distributed across regions and identify stronger and weaker regional performance.

This provides useful information for evaluating geographical business performance.

8.7 Sales Distribution by Customer Segment

A donut chart was created to show the distribution of sales across customer segments.

The dashboard includes the following segments:

- **Consumer**
- **Corporate**
- **Home Office**

This visualization provides a quick understanding of the contribution of different customer segments to overall sales.

8.8 Interactive Filters

Two interactive slicers were incorporated into the dashboard:

Year Filter

Allows users to select different years and dynamically update the dashboard visuals.

Region Filter

Allows users to filter the dashboard based on geographical region.

The slicers were tested to ensure that the relevant KPI cards and visualizations respond correctly to user selections.

8.9 Dashboard Design and Formatting

The dashboard was designed with a clean and professional business-oriented layout.

The following design principles were applied:

- Consistent blue and white color theme.
- Clear dashboard title and subtitle.
- Consistent KPI card formatting.
- Readable chart titles.
- Organized visual placement.
- Consistent slicer formatting.
- Appropriate spacing between visual elements.
- Interactive filtering for improved usability.

The dashboard was designed to provide important information on a **single page**, allowing users to quickly understand overall **performance** and explore different business dimensions.

8.10 Final Power BI Dashboard

The final Power BI dashboard provides an interactive one-page view of the financial performance of the Superstore sample dataset. It combines key performance indicators, category analysis, product performance, monthly sales trends, regional analysis, customer-segment analysis, and interactive Year and Region filters.



Financial Performance Analysis Dashboard

9. KEY BUSINESS INSIGHTS

1. Overall Sales Performance

The dataset generated total sales of approximately 2.33 million, with total profit of approximately 292.30K.

2. Category Performance

Technology generated the highest sales among the three categories, followed by Furniture and Office Supplies.

3. Regional Performance

The West region recorded the highest sales, followed by the East, Central, and South regions.

4. Segment Performance

The Consumer segment contributed the largest share of sales, followed by Corporate and Home Office.

5. Monthly Sales Trend

Monthly analysis showed variations in sales throughout the years, helping identify periods of relatively higher and lower sales performance.

6. Product Performance

The Top 10 Product analysis helped identify products contributing significantly to overall sales.

10. BUSINESS RECOMMENDATIONS

Based on the analysis performed using Excel, SQL, and Power BI, the following recommendations can be considered:

10.1 Focus on High-Performing Products

Products with consistently high sales can be monitored closely to maintain their availability and support continued revenue generation.

10.2 Monitor Profitability Along with Sales

Sales alone should not be used to evaluate performance. Products and categories should also be assessed based on their profitability to support better business decisions.

10.3 Analyze Regional Performance

Differences in regional performance can help identify areas that require additional attention and areas that are performing strongly.

10.4 Understand Customer Segments

Analyzing Consumer, Corporate, and Home Office segments can help businesses understand their major customer groups and develop more targeted strategies.

10.5 Monitor Monthly Trends

Regular monitoring of monthly sales trends can help identify changes in performance and support better planning and forecasting.

10.6 Use Interactive Reporting

An interactive dashboard can make business reporting more efficient by allowing users to filter information by year and region instead of manually analyzing the entire dataset.

11. CONCLUSION

The **Financial Performance Analysis** project demonstrates an end-to-end data analytics workflow using **Excel, SQL, and Power BI**.

Excel was used for data preparation and initial analysis, while SQL was used to perform structured queries and extract business insights from the transaction data. Power BI was then used to transform the results into an interactive dashboard containing KPIs, charts, and filters.

The project provided experience in working with transaction-level data, performing business-oriented analysis, creating analytical measures, and communicating insights through data visualization.

Overall, the project demonstrates how multiple data analytics tools can be combined to transform raw data into meaningful information that can support **data-driven business decision-making**.